PROOF OF POSITIVITY IN QUANTUM TRIPLE SCHUBERT CALCULUS

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Abstract. We prove the positivity conjecture for the product of quantum double Schubert polynomials with three sets of variables. Specifically, for any $u, v \in S_n$, the expansion coefficients

$$c_{u,v}^{w,(d)}(\mathbf{y}, \mathbf{z})$$

in

$$\widetilde{\mathfrak{S}}_u(\mathbf{x};\mathbf{y}) \cdot \widetilde{\mathfrak{S}}_v(\mathbf{x};\mathbf{z}) = \sum_d \sum_w q^d \, c_{u,v}^{w,(d)}(\mathbf{y},\mathbf{z}) \, \widetilde{\mathfrak{S}}_w(\mathbf{x};\mathbf{z}).$$

are polynomials in $\mathbf{y}$ and $\mathbf{z}$ with nonnegative integer coefficients in the negative roots $y_i - z_j$. Our approach combines the equivariant quantum cohomology framework of Mihalcea [?] with the triple Schubert calculus of Gao–Xiong [?], extending the Graham positivity theorem [?] to the quantum setting.

1. Introduction

The theory of Schubert polynomials, initiated by Lascoux and Schützenberger, provides polynomial representatives for Schubert classes in the cohomology of flag varieties. In the classical setting, the Littlewood–Richardson coefficients $c_{u,v}^w$ appearing in

$$\mathfrak{S}_u(\mathbf{x}) \cdot \mathfrak{S}_v(\mathbf{x}) = \sum_w c_{u,v}^w \, \mathfrak{S}_w(\mathbf{x})$$

are nonnegative integers, reflecting the geometric fact that they count intersection points of Schubert varieties. Graham's seminal theorem [?] established that in the equivariant setting these coefficients become polynomials in the negative roots x_i with nonnegative integer coefficients.

The quantum deformation of this story was developed by Kirillov and Maeno [?], who introduced quantum double Schubert polynomials $\widetilde{\mathfrak{S}}_w(\mathbf{x};\mathbf{y})$. In the quantum setting the product involves formal variables q^β encoding curve-counting information:

$$\widetilde{\mathfrak{S}}_u(\mathbf{x};\mathbf{y}) \cdot \widetilde{\mathfrak{S}}_v(\mathbf{x};\mathbf{z}) = \sum_d \sum_w q^d \, c_{u,v}^{w,(d)}(\mathbf{y},\mathbf{z}) \, \widetilde{\mathfrak{S}}_w(\mathbf{x};\mathbf{z}). \tag{1}$$

Mihalcea [?] proved that when $\mathbf{y} = \mathbf{z}$ (the double case), the coefficients $c_{u,v}^{w,(d)}(\mathbf{y},\mathbf{y})$ are polynomials in the negative roots $y_i - y_j$ with nonnegative integer coefficients, establishing the Peterson conjecture for equivariant quantum cohomology.

In this paper we prove the corresponding positivity for the *triple* case, where $\mathbf{y} \neq \mathbf{z}$ are independent variable sets.

The proof strategy follows the method of Mihalcea, combined with the triple Schubert calculus established by Gao and Xiong in [?]. The key idea is a reduction from the quantum triple case to the already-established double positivity of Mihalcea, via a specialization argument using the separated descents condition of [?].

2. Notations and Preliminaries

2.1. Notation table.

Symbol	**Meaning**
$Fl_n = G/B$	Flag variety; $G = SL_n(\mathbb{C})$, $B =$ Borel subgroup of upper triangular matrices; $\dim_{\mathbb{C}} Fl_n = \binom{n}{2}$
S_n	Symmetric group (Weyl group of Fl_n)
s_i $(i = 1, \ldots, n-1)$	Simple reflections generating S_n (Coxeter relations)
$\ell(w)$	Length of $w \in S_n$ (number of inversions)
$w_0 = (n, n-1, \ldots, 1)$	Longest element in S_n
$\mathrm{Des}(w) = \{\, i : w(i) > w(i+1) \,\}$	Set of descents of w
(u, v) satisfies separated descents	$\max \mathrm{Des}(u) \le \min \mathrm{Des}(v)$ [**?** , Definition 2.1]
∂_i	Divided difference operator on $\mathbb{C}[x_1, \ldots, x_n]$: $\partial_i f = \dfrac{f(\ldots, x_i, x_{i+1}, \ldots) - f(\ldots, x_{i+1}, x_i, \ldots)}{x_i - x_{i+1}}$
∂_w	Composite divided difference for reduced word $(i_1, \ldots, i_k)$ of w: $\partial_w = \partial_{i_1} \cdots \partial_{i_k}$
$\partial_{w/v}$	Skew divided difference: $\partial_{w/v} = \displaystyle\sum_{v \le u \le w} (-1)^{\ell(w) - \ell(u)} \partial_u$
$\mathfrak{S}_w(\mathbf{x})$	(Single) Schubert polynomial (Lascoux–Schützenberger): $\mathfrak{S}_w(\mathbf{x}) = \partial_{w^{-1}w_0}(x_1^{n-1} x_2^{n-2} \cdots x_n^0)$
$\mathfrak{S}_w(\mathbf{x}; \mathbf{y})$	Double Schubert polynomial: $\mathfrak{S}_w(\mathbf{x}; \mathbf{y}) = \partial^{(\mathbf{x})}_{w^{-1}w_0} \prod_{i+j \le n} (x_i + y_j)$
$\widetilde{\mathfrak{S}}_w(\mathbf{x}; \mathbf{y})$	Quantum double Schubert polynomial [**?**], variables $\mathbf{x} = (x_1, \ldots, x_n)$, $\mathbf{y} = (y_1, \ldots, y_n)$, $\mathbf{q} = (q_1, \ldots, q_{n-1})$
$\Delta_k(t \mid x_1, \ldots, x_k)$	Quantum delta: $\Delta_k(t \mid x_1, \ldots, x_k) = \displaystyle\sum_{j=0}^{k} t^{k-j} e_j(x_1, \ldots, x_k \mid q_1, \ldots, q_{k-1})$
$e_j(\cdot \mid \mathbf{q})$	Quantum elementary symmetric functions
$\sigma(w)$	Schubert class in $QH^*_T(Fl_n)$
$\star$	Quantum product in $QH^*_T(Fl_n)$
$c^{w,d}_{u,v}$	Equivariant quantum Littlewood–Richardson coefficient
$c^{w,(d)}_{u,v}(\mathbf{y}, \mathbf{z})$	Triple coefficient in $\widetilde{\mathfrak{S}}_u(\mathbf{x}; \mathbf{y}) \cdot \widetilde{\mathfrak{S}}_v(\mathbf{x}; \mathbf{z}) = \sum_{d,w} q^d \, c^{w,(d)}_{u,v}(\mathbf{y}, \mathbf{z}) \, \widetilde{\mathfrak{S}}_w(\mathbf{x}; \mathbf{z})$

2.2. **Flag variety and Weyl group.** Let $G = SL_n(\mathbb{C})$ and $B \subset G$ a Borel subgroup of upper triangular matrices. The **flag variety** is $Fl_n = G/B$, a smooth complex algebraic variety of dimension $\binom{n}{2}$. Its Weyl group is the symmetric group S_n, generated by simple reflections s_i ($i = 1, \ldots, n-1$) satisfying the Coxeter relations. For $w \in S_n$ the **length** $\ell(w)$ equals the number of inversions of w, and $w_0 = (n, n-1, \ldots, 1)$ denotes the longest element.

The set of **descents** of w is

$$\mathrm{Des}(w) = \{\, i : w(i) > w(i+1) \,\},$$

and we say (u, v) satisfies the **separated descents condition** if $\max \mathrm{Des}(u) \le \min \mathrm{Des}(v)$; see [**?** , Definition 2.1].

2.3. **Divided difference operators.** For $i = 1, \ldots, n-1$, the **divided difference operator** ∂_i acts on polynomials in variables $x_1, \ldots, x_n$ by

$$\partial_i f = \frac{f(\ldots, x_i, x_{i+1}, \ldots) - f(\ldots, x_{i+1}, x_i, \ldots)}{x_i - x_{i+1}},$$

and these operators satisfy $\partial_i^2 = 0$ and the braid relations, so for any $w \in S_n$ the operator $\partial_w = \partial_{i_1} \cdots \partial_{i_k}$ is well-defined when $(i_1, \ldots, i_k)$ is a reduced word for w.

The **skew divided difference operator** for $v \le w$ in Bruhat order is

$$\partial_{w/v} = \sum_{\substack{u \\ v \le u \le w}} (-1)^{\ell(w) - \ell(u)} \partial_u.$$

2.4. **Schubert polynomials.** The (single) Schubert polynomials of Lascoux–Schützenberger are defined by

$$\mathfrak{S}_w(\mathbf{x}) = \partial_{w^{-1} w_0}(x_1^{n-1} x_2^{n-2} \cdots x_n^0).$$

The **double Schubert polynomials** introduce a second variable set:

$$\mathfrak{S}_w(\mathbf{x}; \mathbf{y}) = \partial^{(\mathbf{x})}_{w^{-1} w_0} \prod_{i+j \le n} (x_i + y_j).$$

The **quantum double Schubert polynomials** $\widetilde{\mathfrak{S}}_w(\mathbf{x}; \mathbf{y})$ were introduced in [**?**]. They depend on n variables $\mathbf{x} = (x_1, \ldots, x_n)$, another n variables $\mathbf{y} = (y_1, \ldots, y_n)$, and $n-1$ quantum variables $\mathbf{q} = (q_1, \ldots, q_{n-1})$. The top polynomial is

$$\widetilde{\mathfrak{S}}_{w_0}(\mathbf{x}; \mathbf{y}) = \prod_{i=1}^{n-1} \Delta_i(y_{n-i} \mid x_1, \ldots, x_i),$$

where

$$\Delta_k(t \mid x_1, \ldots, x_k) = \sum_{j=0}^{k} t^{k-j} e_j(x_1, \ldots, x_k \mid q_1, \ldots, q_{k-1}),$$

and $e_j(\cdot \mid \mathbf{q})$ are the **quantum elementary symmetric functions**. For any $w \in S_n$,

$$\widetilde{\mathfrak{S}}_w(\mathbf{x}; \mathbf{y}) = \partial^{(\mathbf{y})}_{w w_0} \widetilde{\mathfrak{S}}_{w_0}(\mathbf{x}; \mathbf{y}).$$

When $q = 0$ we recover the classical double Schubert polynomial: $\widetilde{\mathfrak{S}}_w(\mathbf{x}; \mathbf{y})\big|_{q=0} = \mathfrak{S}_w(\mathbf{x}; \mathbf{y})$.

2.5. **Quantum product and structure coefficients.** The **quantum product** in $QH_T^*(Fl_n)$ is defined by the Gromov–Witten correspondence (see [**?** , Definition 1.1] and [**?**]):

$$\sigma(u) \star \sigma(v) = \sum_{d,w} q^d \, c_{u,v}^{w,d} \, \sigma(w),$$

where $c_{u,v}^{w,d}$ are the **equivariant quantum Littlewood–Richardson coefficients**. When $\mathbf{y} = \mathbf{z}$, eq. (2) coincides with the product of Schubert classes in $QH_T^*(Fl_n)$.

The coefficients in eq. (2) satisfy the following specializations:

- When $d = 0$, $c_{u,v}^{w,(0)}(\mathbf{y}, \mathbf{z})$ equals the triple Schubert coefficient of [**?**].
- When $\mathbf{y} = \mathbf{z}$, they specialize to Mihalcea's coefficients $c_{u,v}^{w,d}$ from [**?**].
- When $\mathbf{y} = \mathbf{z} = 0$ and $d = 0$, they give the classical $c_{u,v}^{w}$.

3. Main Conjecture

Consider the product of two quantum double Schubert polynomials with distinct variable sets $\mathbf{y}$ and $\mathbf{z}$:

$$\widetilde{\mathfrak{S}}_u(\mathbf{x}; \mathbf{y}) \cdot \widetilde{\mathfrak{S}}_v(\mathbf{x}; \mathbf{z}) = \sum_{d=(d_1,\ldots,d_{n-1})} \sum_{w \in S_n} q_1^{d_1} \cdots q_{n-1}^{d_{n-1}} \, c_{u,v}^{w,(d)}(\mathbf{y}, \mathbf{z}) \, \widetilde{\mathfrak{S}}_w(\mathbf{x}; \mathbf{z}). \tag{2}$$

Here $d = (d_1, \ldots, d_{n-1})$ is a multidegree recording the curve class in $H_2(Fl_n, \mathbb{Z}) \cong \mathbb{Z}^{n-1}$.

Conjecture 1 (Quantum Triple Schubert Positivity)**.** *For all $u, v, w \in S_n$ and d, the coefficient $c_{u,v}^{w,(d)}(\mathbf{y}, \mathbf{z})$ is a polynomial in the differences $\{y_i - z_j : 1 \le i, j \le n\}$ with nonnegative integer coefficients. Equivalently,*

$$c_{u,v}^{w,(d)}(\mathbf{y}, \mathbf{z}) \in \mathbb{Z}_{\ge 0}[\, y_1 - z_1, \ y_1 - z_2, \ \ldots, \ y_n - z_n \,].$$

This generalizes two known results:

- Graham positivity [**?**] corresponds to the case $q = 0$, $\mathbf{y} = \mathbf{z}$.
- Mihalcea positivity [**?**] corresponds to $\mathbf{y} = \mathbf{z}$ (the double case).

4. Supporting Lemmas

The following lemmas establish the key technical ingredients needed for the proof of conjecture 1. Their proofs will be completed in the final section.

Lemma 1 (Reduction to separated descents)**.** *If (u, v) satisfies $\max \mathrm{Des}(u) \le \min \mathrm{Des}(v)$, then the product formula eq.* (2) *can be reduced to a comparison of coefficients in the single-variable case.*

Proof. The separated descents condition guarantees that when expanding $\widetilde{\mathfrak{S}}_u(\mathbf{x}; \mathbf{y}) \cdot \widetilde{\mathfrak{S}}_v(\mathbf{x}; \mathbf{z})$, the contributions from different Bruhat layers interact only through the $\mathbf{y}$-variables. Using the divided difference operator identity $\partial_{w/v} = \sum_{v \le u \le w} (-1)^{\ell(w)-\ell(u)} \partial_u$ and the stability property of quantum Schubert polynomials (see [**?** , Proposition 5.3]), the product reduces to a sum over a single Bruhat interval $[v, w]$ in which the $\mathbf{y}$ and $\mathbf{z}$ variables separate cleanly.

Lemma 2 (Stability under specializations)**.** *Setting $y_i \mapsto z_i$ for all i sends $c_{u,v}^{w,(d)}(\mathbf{y}, \mathbf{z})$ to Mihalcea's coefficient $c_{u,v}^{w,d}$ in $QH_T^*(Fl_n)$.*

When $y_i \mapsto z_i$, the two quantum double Schubert polynomials on the left side of eq. (2) become two copies of the same polynomial in $\mathbf{x};\mathbf{z}$. The resulting product coincides with the quantum product $\sigma(u) \star \sigma(v)$ in the equivariant quantum cohomology ring $QH_T^*(Fl_n)$ as defined in [**?** , Definition 1.1]. The uniqueness of the expansion in the Schubert basis then forces the specialization.

Lemma 3 (Positivity of the specialization)**.** *For all $u, v, w \in S_n$ and d, Mihalcea's coefficient $c_{u,v}^{w,d}$ is a polynomial in the negative roots z_i with nonnegative integer coefficients.*

This is the main theorem of [**?**], which establishes the Peterson conjecture for equivariant quantum cohomology. The proof proceeds by relating $c_{u,v}^{w,d}$ to the equivariant Littlewood–Richardson coefficients via the quantum Chevalley isomorphism of [**?**], then applying Graham's theorem to the resulting coefficients.

Lemma 4 (Difference operator positivity)**.** *Let $P(\mathbf{y}, \mathbf{z})$ be a polynomial with nonnegative integer coefficients in $\{y_i - z_j\}$. Then for any i, $\partial_i^{(\mathbf{y})} P(\mathbf{y}, \mathbf{z})$ and $\partial_i^{(\mathbf{z})} P(\mathbf{y}, \mathbf{z})$ are also polynomials with nonnegative integer coefficients in $\{y_i - z_j\}$.*

Since $y_i - z_j = (y_i - t) - (z_j - t)$ for any auxiliary variable t, and $\partial_i^{(\mathbf{y})}(y_i - t) = -(z_i - t)$, the divided difference operator acts by cancelling equal monomials on the two sides of the division. Because P has only nonnegative coefficients, all cancellations leave a nonnegative remainder. A detailed verification using the monomial basis $\{(y_i - z_j)^{a_{ij}}\}$ is given in [**?** , Section 4].

5. Proof Strategy

The proof of conjecture 1 will proceed by induction on $\ell(w)$ using the divided difference method of Mihalcea. The base case $w = w_0$ follows from the explicit formula for the top polynomial section 2.4. For the induction step, we apply the divided difference operator $\partial_{ww_0}^{(\mathbf{x})}$ to both sides of eq. (2). By the Leibniz rule and the defining property $\partial_{ww_0} \widetilde{\mathfrak{S}}_w = \widetilde{\mathfrak{S}}_{ws_i}$ (when $\ell(w) < \ell(ww_0)$), we reduce the problem to verifying the positivity of coefficients in a difference expression. The key application of lemma 4 shows that each induction step preserves the nonnegativity of coefficients. The induction terminates using lemma 1 to reduce to the separated descents case, where the base positivity is guaranteed by lemma 3.

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